



Metformin

Generic name: metformin [*met-FOR-min*]

Brand names: Fortamet (discontinued), Glucophage (discontinued), Glucophage XR (discontinued), Glumetza, Riomet

Drug class: Biguanides

Medically reviewed by Melisa Puckey, BPharm. Last updated on Oct 1, 2026.

What is metformin?

Metformin is a first-line treatment used for type 2 diabetes in adults and children 10 years and older to help improve blood sugar levels. It is also used off-label to improve symptoms of polycystic ovarian syndrome (PCOS) and other conditions. Metformin tablets should be used together with dietary changes and exercise for better results.

Metformin belongs to the drug class biguanides.

Metformin works by reducing glucose absorption from the intestines, lowering liver glucose production, and improving insulin sensitivity. Managing blood sugar levels with medications like metformin can prevent diabetes complications such as kidney damage, nerve issues, blindness, amputations, sexual dysfunction, and lower the risk of cardiovascular disease.

Warnings

Boxed warning for lactic acidosis

Metformin has a boxed warning for lactic acidosis, a dangerous build-up of lactic acid in your blood, which can cause death, hypothermia, hypotension, and resistant bradyarrhythmias. **A boxed warning is the strongest safety-related warning issued by the FDA.**

This is an extremely rare side effect but certain things can raise your risk of lactic acidosis, including kidney problems, taking certain other medicines, being over 65, having an imaging test that uses contrast dye, having surgery or other medical procedures, conditions that lower oxygen levels in the body, drinking too much alcohol, liver problems, and certain rare inherited metabolic disorders (mitochondrial diseases).

Stop taking metformin and call your healthcare professional immediately if you have any symptoms of lactic acidosis, including if you feel:

- very weak and tired
- have unusual (not normal) muscle pain
- have trouble breathing
- have unusual sleepiness or sleep longer than usual
- have unexplained stomach or intestinal problems with nausea and vomiting, or diarrhea
- cold, especially in your arms and legs
- dizzy or lightheaded
- have a slow or irregular heartbeat

Other warnings

Vitamin B12 deficiency: Metformin can lower your vitamin B12 levels over time. Your doctor may check your blood counts every year and your B12 levels every 2 to 3 years to catch any problems early.

Low blood sugar when taken with insulin or other diabetes medicines: Taking metformin along with insulin or certain other diabetes drugs (ones that boost insulin release) raises the risk of low blood sugar. If you're on one of these combinations, your doctor may lower the dose of the other medicine.

Macrovascular Outcomes: There have been no clinical studies that clearly show metformin reduces the risk of macrovascular disease, such as diseases or conditions that affect the coronary arteries, the aorta, and large arteries in the brain and limbs.

Metformin side effects

Common side effects

Common side effects of metformin hydrochloride tablets include diarrhea, nausea, vomiting, wind (flatulence), indigestion, and headache. These side effects occurred in 5% or more of patients in the US clinical trial of using metformin for type 2 diabetic patients. In the same study, 6% of patients discontinued metformin due to diarrhea.

Less common side effects were abnormal stools, hypoglycemia, myalgia, lightheadedness, dyspnea, nail disorder, rash, increased sweating, taste disorder, chest discomfort, chills, flu syndrome, flushing, and palpitation, which occurred between 1% and less than 5% of patients.

Taking your medicine with meals can help reduce these side effects, and you may find the slow-release tablets reduce gastric side effects. Tell your doctor if the side effects bother you a lot, last for more than a few weeks, come back after they've gone away, or start later in therapy. You may need a lower dose or need to stop taking the medicine for a short period or for good.

Metformin rarely causes hypoglycemia (low blood sugar) by itself. However, hypoglycemia can happen if you do not eat enough, if you drink alcohol, or if you take other medicines to lower blood sugar.

Serious side effects

Get emergency medical help if you have signs of an allergic reaction to metformin: hives; difficult breathing; swelling of your face, lips, tongue, or throat.

This medicine can rarely cause patients to develop lactic acidosis, which can be fatal. See the "Warnings - Boxed Warning" for more details.

Call your doctor at once if you have signs of liver problems, such as upper stomach pain, loss of appetite, dark urine, clay-colored stools, or yellowing of your skin or eyes (jaundice).

Contraindications

You should not use metformin if you have:

- severe kidney disease, or
- metabolic acidosis or diabetic ketoacidosis (call your doctor for treatment), or
- an allergy to metformin or any of the inactive ingredients in this medicine.

Before taking this medicine

If you need to have surgery or any type of x-ray or CT scan using a dye that is injected into your veins, you may need to temporarily stop taking metformin. Be sure your caregivers know ahead of time that you are using this medication.

Tell your doctor if you have ever had:

- kidney disease (your kidney function may need to be checked before you take this medicine);
- high ketone levels in your blood or urine;
- heart disease, congestive heart failure;
- liver disease or liver impairment
- if you also use insulin, or other oral diabetes medications.

You may develop lactic acidosis, a dangerous build-up of lactic acid in your blood. This may be more likely if you have other medical conditions, a severe infection, chronic alcoholism, or if you are 65 or older. Ask your doctor about your risk.

Metformin immediate release tablets should not be given to a child younger than 10 years old.

This medicine should not be used for type 1 diabetes.

Pregnancy

Follow your doctor's instructions about using metformin if you are pregnant or you become pregnant. Controlling diabetes is very important during pregnancy, and having high blood sugar may cause complications in both the mother and the baby. Tell your doctor if you become pregnant while taking metformin.

Metformin may stimulate ovulation in some women and may increase the risk of unintended pregnancy. Talk to your doctor about your risk.

Breastfeeding

Talk to your healthcare provider if you are breastfeeding or plan to breastfeed. Metformin can pass into your breast milk. Your healthcare provider will be able to advise you about the best way to feed your baby while you take this medicine.

How should I take metformin?

Take metformin exactly as prescribed by your doctor. Follow all directions on your prescription label and read all medication guides or instruction sheets. Your doctor may occasionally change your dose. Use the medicine exactly as directed.

Take metformin with a meal, unless your doctor tells you otherwise. Some forms of metformin are taken only once daily with the evening meal. Follow your doctor's instructions.

Take immediate release tablets in divided doses 2 to 3 times a day with meals. Your dose will be increased slowly to minimise gastrointestinal side effects.

Do not crush, chew, or break an extended-release tablet. Swallow it whole.

Measure liquid medicine carefully. Shake the oral suspension before you measure a dose. Use the dosing syringe provided, or use a medicine dose-measuring device (not a kitchen spoon).

Some tablets are made with a shell that is not absorbed or melted in the body. Part of this shell may appear in your stool. This is normal and will not make the medicine less effective.

You may have low blood sugar (hypoglycemia) and feel very hungry, dizzy, irritable, confused, anxious, or shaky. To quickly treat hypoglycemia, eat or drink a fast-acting source of sugar (fruit juice, hard candy, crackers, raisins, or non-diet soda).

Your doctor may prescribe a glucagon injection kit in case you have severe hypoglycemia. Be sure your family or close friends know how to give you this injection in an emergency.

Blood sugar levels can be affected by stress, illness, surgery, exercise, alcohol use, or skipping meals. Ask your doctor before changing your dose or medication schedule.

Metformin is only part of a complete treatment program that may also include diet, exercise, weight control, regular blood sugar testing, and special medical care. Follow your doctor's instructions very closely.

Store at room temperature away from moisture, heat, and light.

Your doctor may have you take extra vitamin B12 while you are taking this medicine. Take only the amount of vitamin B12 that your doctor has prescribed.

Metformin dose Information

Usual Adult Metformin Dose for Diabetes Type 2

Immediate-release tablets

Initial dose: 500 mg orally twice a day or 850 mg orally once a day

Dose titration: Increase by 500 mg increments weekly or 850 mg every 2 weeks as tolerated.

Maintenance dose: 2000 mg/day in divided doses

Maximum dose: 2550 mg/day. Doses above 2,000 mg may be better tolerated given 3 times

Extended-release tablets

Initial dose: 500 to 1000 mg orally once a day

Dose titration: Increase in 500 mg increments weekly as tolerated

Maximum dose: 2000 mg/day

Usual Pediatric Dose for Diabetes Type 2 (10 years or older)

Immediate-release tablets

Initial dose: 500 mg orally twice a day

Dose titration: Increase in 500 mg increments weekly as tolerated; daily dose should be taken in divided doses 2 to 3 times a day with meals

Maximum dose: 2000 mg/day

Extended-release oral suspension

Initial dose: 500 mg orally once a day with evening meal

Dose titration: Increase in 500 mg increments weekly based on glycemic control and tolerability

Maximum dose: 2000 mg/day

Renal Impairment

- Before starting treatment, assess renal function with estimated glomerular filtration rate (eGFR)
- Do **not** use this medicine in patients with eGFR below 30 mL/minute/1.73 m²
- It is not recommended to start metformin tablets in patients with eGFR between 30 to 45 mL/minute/1.73 m²
- While taking this medicine, assess the risk/benefit of continuing if eGFR falls below 45 mL/minute/1.73 m²
- If eGFR falls below 30 mL/minute/1.73 m² discontinue this medication

Metformin immediate release tablets are available as 500 mg, 850 mg, and 1,000 mg tablets.

Hepatic Impairment

Avoid use in patients with hepatic impairment.

What happens if I miss a dose?

Take the medicine as soon as you can, but skip the missed dose if it is almost time for your next dose. Do not take two doses at one time.

What happens if I overdose?

Seek emergency medical attention or call the Poison Help line at 1-800-222-1222. An overdose can cause severe hypoglycemia or lactic acidosis.

What to avoid

Avoid drinking alcohol. It lowers blood sugar and may increase your risk of lactic acidosis.

What other drugs will affect metformin?

Many drugs can interact with metformin; this includes prescription and over-the-counter medicines, vitamins, and herbal products. Not all possible interactions are listed here. Tell your doctor about all your current medicines and any medicine you start or stop using.

Medicines that have a carbonic anhydrase inhibiting effect like topiramate (often used for seizures or migraines), zonisamide (used for seizures), acetazolamide (used for glaucoma, altitude sickness, and other conditions), and dichlorphenamide (used for a muscle condition called periodic paralysis) can raise the risk of a rare but serious condition called lactic acidosis when taken with metformin. These medicines commonly lower a substance in your blood called bicarbonate, which your body needs to keep its acid levels balanced. If you're taking metformin along with any of these medicines, your doctor may want to monitor you more closely.

Other medicines that may increase risk of lactic acidosis. Some medicines including cimetidine (used for heartburn), ranolazine (used for chest pain/angina), dolutegravir (used to treat HIV), and vandetanib (used to treat certain thyroid cancers) can all cause metformin levels to increase. They do this by blocking the same pathway your kidneys use to clear metformin out of your system. Since higher metformin levels raise the risk of lactic acidosis, your doctor will weigh the benefits and risks before prescribing any of these together with metformin.

Insulin or other diabetes medicines that raise insulin (like sulfonylureas): Combining these with metformin can lower your blood sugar too much. Your doctor may need to adjust your dose if you're taking more than one diabetes medicine.

Alcohol: Drinking alcohol, especially in large amounts or regularly, can make it harder for your body to process metformin normally and raises the risk of lactic acidosis.

Contrast dye used in imaging tests (like CT scans): This type of dye can temporarily affect kidney function, which can let metformin build up in your body. Your doctor may have you stop taking metformin around the time of the test and restart it once your kidney function is confirmed to be back to normal.

Metformin for Weight Loss: What the Research Shows

Does metformin work for weight loss?

Yes. Metformin can lead to a modest weight loss, though results vary, and it's generally smaller amount of weight loss than what's seen with medications specifically approved for weight loss.

In the landmark Diabetes Prevention Program (DPP), a randomized controlled trial in people at high risk for diabetes, participants taking metformin lost more weight than those on placebo.

Metformin patients lost 2.7% of body weight after one year, compared to 0.43% for placebo, and 2.1% after two years compared to essentially no change (0.02%) for placebo. Looking at it another way, **29% of people on metformin lost at least 5% of their starting body weight after one year, versus just 13% on placebo**, a gap that held up at the two-year mark (26% vs. 14%).

Metformin is not FDA-approved for weight loss.

How does metformin help PCOS?

Metformin is prescribed to manage polycystic ovary syndrome (PCOS) because it improves how the body responds to insulin. PCOS affects about 1 in 10 women of reproductive age and is often marked by insulin resistance, in which the body doesn't respond well to insulin, prompting the pancreas to produce more. This excess insulin drives up androgen (male hormone) production, worsening symptoms like irregular periods, infertility, acne, and excess hair growth. Metformin helps by lowering insulin levels, decreasing the liver's glucose production, improving glucose uptake by muscles, and reducing circulating androgen levels as well, which can support more regular ovulation and menstrual cycles.

Although metformin is not FDA-approved to treat PCOS, it's widely prescribed as an off-label treatment, particularly for those with insulin resistance or fertility difficulties, and it's backed by extensive research and clinical guidelines. Beyond helping regulate cycles and support ovulation, metformin may also help reduce androgen-related symptoms such as acne and excess hair growth, and can aid with weight management for those affected by PCOS-related weight gain.

References

1. Saadati S, Mason T, Godini R, Vanky E, Teede H, Mousa A. Metformin use in women with polycystic ovary syndrome (PCOS): Opportunities, benefits, and clinical challenges. *Diabetes Obes Metab*. 2025;27(Suppl 3):31-47. doi:10.1111/dom.16422
2. Teede HJ, et al.; International PCOS Network. International Evidence-based Guideline for the Assessment and Management of Polycystic Ovary Syndrome 2023. Monash University, Monash Centre for Health Research and Implementation. Accessed July 17, 2026. <https://www.monash.edu/medicine/mchri/pcos/guideline>
3. Attia GM, Almouteri MM, Alnakhli FT. Role of Metformin in Polycystic Ovary Syndrome (PCOS)-Related Infertility. *Cureus*. 2023;15(8):e44493. doi:10.7759/cureus.44493
4. World Health Organization. Polycystic ovary syndrome. Fact sheet. Accessed July 17, 2026. <https://www.who.int/news-room/fact-sheets/detail/polycystic-ovary-syndrome>

Popular FAQ

When is the best time to take metformin?



Can you take metformin and Ozempic together?



How long does metformin take to work?



What is the safest diabetes medication?



Why does metformin cause diarrhea?



Can you take metformin without food?



Does metformin lower A1C, if so how much?



Does metformin cause weight loss?



[What are the symptoms of Metformin related lactic acidosis?](#)



[View more FAQ...](#)

References

1. Riomet Prescribing Information
2. Metformin Prescribing Information
3. Metformin Oral Solution Prescribing Information
4. Metformin Extended Release Tablets Prescribing Information
5. Glumetza Prescribing Information

Further information

Always consult your healthcare provider to ensure the information displayed on this page applies to your personal circumstances.